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INTRODUCTION

Despite certified measles elimination in 2020 [1], Belgium counted the third-highest measles notification rate among European Union and European Economic Area (EU/EEA) countries in 2024 [2].

To maintain elimination status, it is paramount to confirm interruption of transmission chains and absence of endemic circulation.

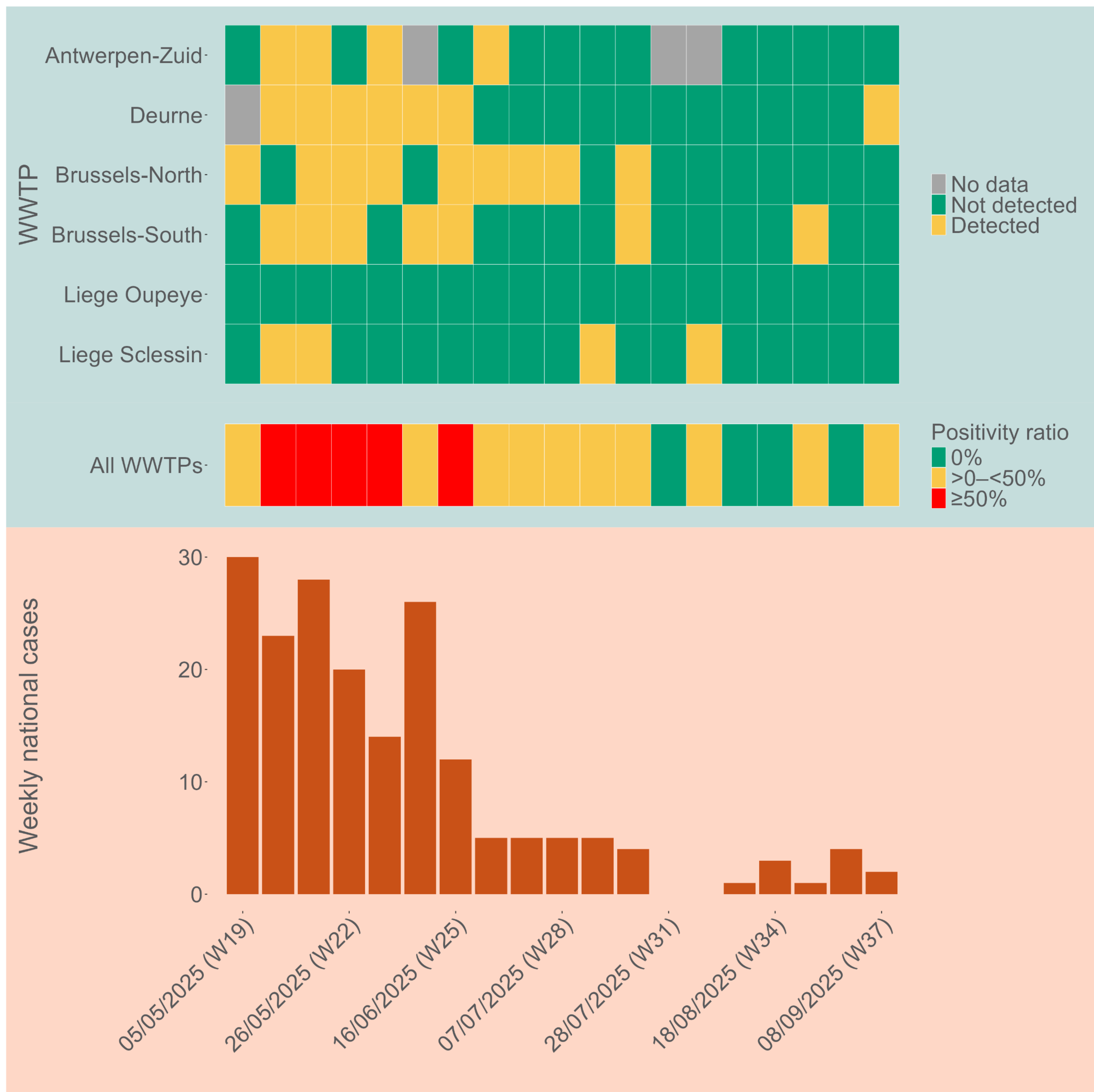
AIM

- Explore the potential of wastewater-based surveillance as a complementary tool to clinical surveillance for monitoring measles.
- Compare detection of measles virus in Belgian wastewater with reported clinical case data.

METHOD

- Wastewater collected from the influents of 6 wastewater treatment plants (WWTPs)
- WWTP selected based on 3 criteria:
 - High population coverage
 - Proximity to identified risk areas
 - Geographic distribution across the Brussels-Capital, Flemish and Walloon regions
- Samples collected weekly May-September 2025
- Detection by RT-dPCR
- Clinical case data retrieved from mandatory notifications to regional authorities and lab testing at the National Reference Centre

RESULT



Note. The depicted clinical cases include possible, probable and confirmed cases.

CONCLUSIONS AND FUTURE STEPS

- Measles virus was successfully detected in Belgian wastewater, and qualitative trends mirrored those observed in clinical surveillance at a national level.
- Further data consolidation is required to conduct a correlation study at the provincial and WWTP level.
- Extend the study to 30 WWTPs covering 38% of the Belgian population.

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